Mollah Adib Aktab

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Engineering Portfolio | LinkedIn | GitHub

Education

Bangladesh University of Engineering & Technology (BUET) 2020 – 2025
Bachelor of Science in Biomedical Engineering *Dhaka, Bangladesh*
CGPA: 3.59/4.00 *(Result published on 25th March, 2025)*

Research Interest

Computational Ophthalmology • Biomechanics • Non-linear Finite Element Analysis • Computational Fluid Dynamics • Simulation-Based Product Design • Product Evaluation

Work Experience

Vanguard Medical 2025 – May 2026
SolidWorks Design and FEA Engineer (Remote) *Birmingham, UK*

- FEA simulations (crashworthiness, static, transient) were performed to validate medical devices against international safety and regulatory standards.
- Stress, deformation, and failure investigated to improve design, safety, and reliability.
- Continuous communication with the manufacturing team in UK until the design properly implemented was an integral part.

BioInnovation Research Group, BUET 2025 – Present
Research Assistant *Dhaka, Bangladesh*

- Prepared project proposals, technical documentation, and research materials to support academic and industry-collaborative projects.
- Managed day-to-day activities in the lab.
- Trained and mentored junior researchers in tools such as ANSYS, COMSOL, and Mimics, fostering team capability and knowledge transfer.

Industrial Product Designing

Smart Connector Design Validation and Optimization for SAE J3027 Testing [Report] 2026
Design and FEA Engineer, Vanguard Medical

- FEA, tolerance analysis, hand calculations, and nonlinear material assessment performed for SmartCot locking and pipe-clamping mechanisms.
- Product behavior was assessed under 750N pull-out force, 22.5g acceleration conditions to identify design limitations and failure risks.
- Geometry and material updated to improve safety, manufacturability, locking reliability and to achieve a test compliant product.

Crashworthiness Analysis and Design Optimization of an ICU/Ambulance Bed for EN 1789 Certification 2025 [Report]
Design and FEA Engineer, Vanguard Medical

- FEA crash simulations of an ICU/ambulance bed under 10g multi-directional loading conducted to ensure compliance with EN 1789 and assess structural performance.
- Stress, deformation, and potential failure zones detected by calculating FOS and take action to improve safety and make the product test compliant.
- Material selection and design needed to reconfigure to enhance strength-to-weight ratio and manufacturability.

Research Experience

Undergraduate Thesis: A Multi-Component Finite Element Model of the Human Eye to Assess Glaucoma Risk under Pathological Conditions 2025 [Report]
Supervisor: Dr. Muhammad Tarik Arafat, Professor and Head, Department of Biomedical Engineering, BUET

- Built a comprehensive 3D eye model in SolidWorks and meshed it with hexahedral elements (ANSYS, FEMAP), and performed node merging to eradicate convergence issue and completed a mesh-convergence study.
- For sclera and Lamina Cribosa, a hyperelastic model was built by curve fitting and validated the behaviour of the model against experimental data. The vitreous body and the enclosed eye fluid were modeled by a hydrostatic fluid element(HSFLD242 implemented by APDL) so that the fluid elements can apply intraocular pressure to

represent the mechanics of the eyes more realistically; the model is validated and benchmarked by published experimental data and eye FE models.
- Ran 4 studies to understand the effect of IOP, ICP, blood pressure and scleral stiffness, and found that IOP is the main cause to damage Laminar Cribosa.

Simulation Lead, DBL Pharmaceuticals Ltd: Comprehensive Studies on the Benefits of Multi-Dose Dry Powder Inhaler (MDPI) [Report] 2025

Supervisor: Dr. Muhammad Tarik Arafat, Professor and Head, Department of Biomedical Engineering, BUET

- Created 3D models of Bexihaler and SykonaHaler in SolidWorks to support simulation-based assessment of multi-dose and single-dose dry powder inhaler devices.
- Performed ANSYS CFD to evaluate and compare airflow distribution, pressure drop and flow rate along with DPM to study particle trajectories, and particle residence time inside the inhaler devices. CFD model was validated by the experimental flow rate.
- Generated comparative engineering insights on MDPI and single-dose inhaler performance to support product assessment, design understanding, and industry-focused device evaluation.

Academic Projects

A Prosthesis Made in Bangladesh for Below Knee Amputees [Report] 2023

- Designed low-cost prostheses using Bézier curve modeling.
- Cost-effective prostheses using locally available materials (Nylon, HDPE) and simple fabrication methods, rigorously tested for safety and durability under load conditions.
- Empowers amputees with restored mobility and dignity by bypassing expensive, complex manufacturing while maintaining clinical functionality.

Glaucoma Detection using Image Processing 2023

- Developed a machine learning-based system for classifying healthy eye and diseased eye.
- Python-based interface for eye health predictions.
- Potential application in detecting glaucoma.

Publications

[1] **Mollah Adib Aktab**[1,*], Marshal Ashif Shawkat, S. M. Mursalin Sonet, Md. Redwan Hossain, and Md. Tazuddin Ahmed, "**Design and Structural Validation of a Low-Cost Prosthetic Ankle-Foot Made from Locally Available Materials**". Submitted to *9th International Conference on Mechanical, Industrial and Energy Engineering*. [Paper]

[2] **Mollah Adib Aktab**, Dr. Mohammed A. Hannan, Dr. Muhammad Tarik Arafat, "**A Multi-Component Finite Element Model of the Human Eye to Assess Glaucoma Risk under Pathological Conditions**". *Manuscript in preparation.*

Skills

Programming: C, Python, MATLAB

3D Modeling: SolidWorks, SpaceClaim, 3-Matic, Mimics

Software: ANSYS (Fluent, Static Structural, Transient, Modal, Rigid Dynamics, Eigenvalue Buckling), COMSOL Multiphysics

Soft Skills: Project Management, Team Collaboration, Communication, Presentation, Problem-Solving

Additional: Microsoft Excel, PowerPoint, Ability to learn and adapt to new areas

Language Test

IELTS – Overall: 7 August 2026

Listening: 7, Reading: 7, Writing: 6.5, Speaking: 7

Training

Industrial Attachment Program for Biomedical Engineer 2024

Beximco Health and PPE, Beximco Industrial Park *Dhaka, Bangladesh*

- Detailed knowledge of manufacturing medical equipment, PPE and masks.
- Acquaintance with professional work environment.

Awards and Honors

Second Prize, Poster Presentation, BME Fest 2024
Second Prize, "Break the Circle" Season XII Photo Exhibition – National Photography Contest 2024
Dean's List Award for Excellent Scholarly Achievement, BUET (Level 1-2) 2021

Leadership

Joint Secretary, BUET Photographic Society 2024
Vice President, BUET Animata 2024
Chair (Event Management Team), ACS 2024

Language

Bangla, English

References

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Dr. Mohammed A. Hannan
Senior Lecturer
Department of Mechanical Engineering, UBC (Vancouver Campus)
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Dr. Jahid Ferdous
Associate Professor
Department of Biomedical Engineering, BUET
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